

BERT — Frozen vs Full Fine-tuning on AG News

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INTRODUCTION

- How much task-specific adaptation does a pretrained language model actually need? Must all weights update, or can frozen representations suffice?

OBJECTIVES

- Reproduce the frozen-encoder vs. full fine-tuning comparison from Devlin et al. (NAACL 2019) on a downstream classification task with controlled, matched conditions.

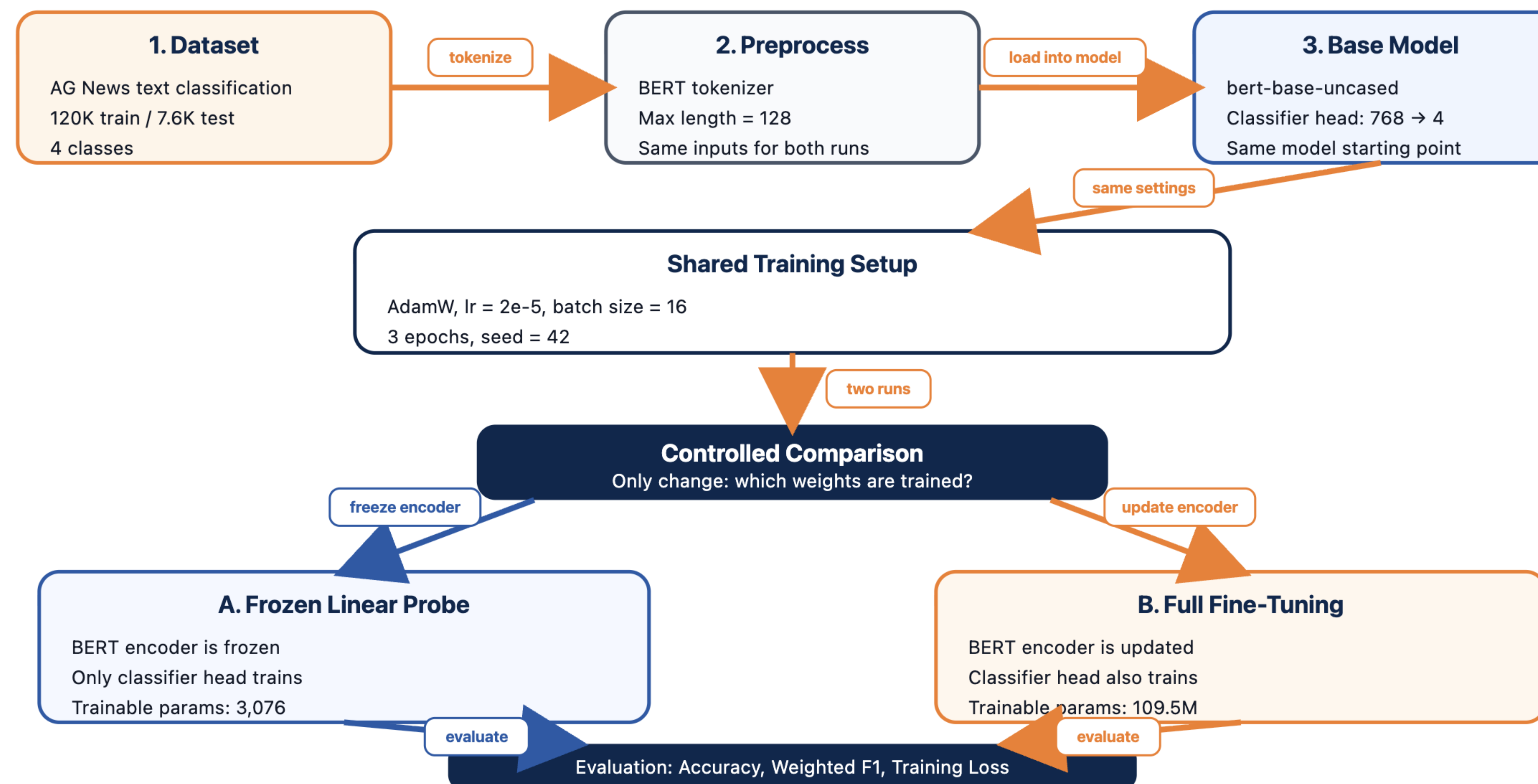
FUTURE WORK

- Partial fine-tuning:** Unfreeze only the top N transformer layers to trace the accuracy vs. compute tradeoff continuously. Find the minimum number of layers needed to recover most of the fine-tuning gain.
- Closer paper replication:** Run the same comparison on SST-2 or MRPC (actual GLUE tasks from Table 1) and with bert-large-uncased to enable a direct numerical comparison.

REFERENCES

- Devlin et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL 2019*.
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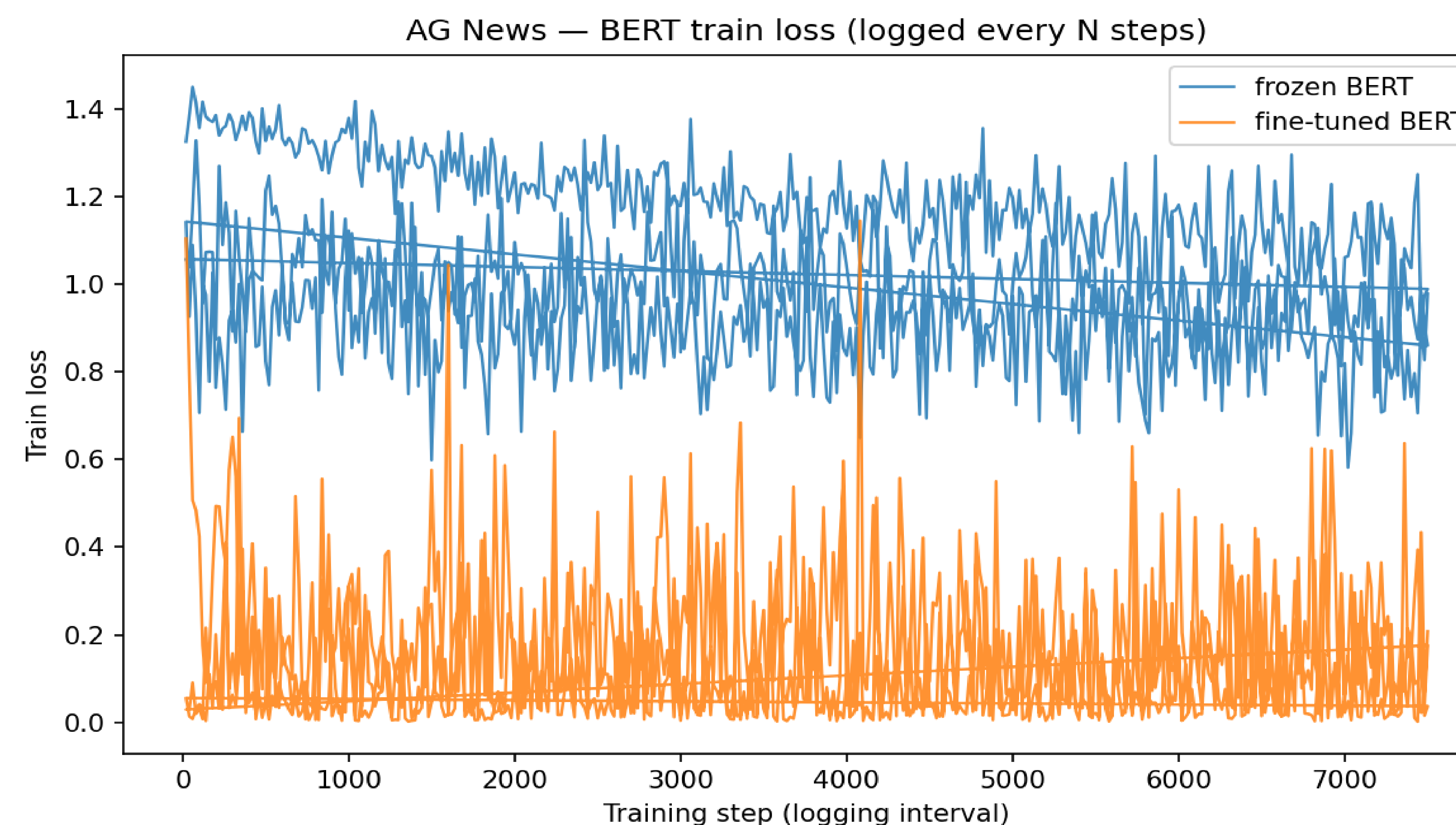
METHODOLOGY



RESULTS

Mode	Test Accuracy	Weighted F1	Trainable Params
Frozen (linear probe)	74.04%	73.87%	3,076
Full fine-tuning	94.63%	94.64%	109,485,316

Table 1. Test-set results after 3 epochs. Full fine-tuning yields a +20.6 pp accuracy gain over the frozen linear probe.



CONCLUSIONS

- Full fine-tuning decisively outperforms linear probing (+20.6 pp), confirming that BERT's frozen representations, while rich, are **not sufficient** for high accuracy on AG News at this scale.
- The loss curves reveal a fundamental behavioral difference: fine-tuning reshapes encoder representations toward the task, while the frozen probe is limited to carving a hyperplane in a fixed 768-D space.
- Within our resource constraints, we reached the same conclusion as **Devlin et al.** even though our dataset, model size, and task differ from the paper's GLUE benchmarks.

Fine-tuning yields a **+20.6 pp accuracy gain** over the frozen probe, replicating the qualitative finding of Devlin et al. Section 5.1.

Loss curves show fine-tuning drops from ~1.1 to near-zero in **epoch 1 alone**; the frozen model descends slowly across all 3 epochs, never reaching those loss levels. Fine-tuning accuracy **saturates after epoch 1** (94.04% → 94.46% → 94.43%), suggesting AG News is close to ceiling at this scale and lr — extra compute adds little.